

DTSA 5510

Unsupervised Algorithms in Machine Learning

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Applied Scientist (50.5 / 100)	46.1 / 100
SPECIALIZATION Machine Learning: Theory and Hands-on Practice with Python		

46

CURRENT AVG SCORE
across 6 job roles

82

POTENTIAL SCORE
with recommended additions

+44

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Students will learn selected unsupervised learning methods for dimensionality reduction, clustering, finding latent features, and application cases such as recommender systems with hands-on examples of product recommendation algorithms. Formerly offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	50.5	95.0	+44.4	HIGH
ML Engineer	50.3	74.8	+24.5	MEDIUM
Machine Learning Engineer	47.4	74.9	+27.6	MEDIUM
AI Engineer	46.6	86.0	+39.4	HIGH
Data Scientist	44.4	86.5	+42.2	HIGH
MLOps Engineer	37.7	72.0	+34.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Business Communication 1 of 6 roles

insights

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

GAP KEYWORDS BY ROLE

Applied Scientist

50.5 → 95.0 (+44.4 pts)

lab models technical data design regulatory science techniques

ML Engineer

50.3 → 74.8 (+24.5 pts)

media data machine models build impact form new

Machine Learning Engineer

47.4 → 74.9 (+27.6 pts)

data interfaces tools design science models applications machine

AI Engineer

46.6 → 86.0 (+39.4 pts)

data models build form technical technology detection machine

Data Scientist

44.4 → 86.5 (+42.2 pts)

data mitigation insights different social build approaches

MLOps Engineer

37.7 → 72.0 (+34.2 pts)

detection data text machine media image tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.